Jacob Salminen

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GitHub: https://github.com/JacobSal
Google Scholar: Jacob Salminen - Google Scholar
Website: https://jacobsal.github.io/

Major Technical Skills	**Software Proficiencies**
Human Research	*Python (Expert)*
Health Professions Education	*MATLAB (Expert)*
Brain & Motion Analyses	*R (Expert)*
Clinical Statistics	*C#/C++/C (Proficient)*
Signal Processing	*GitHub, Docker (Proficient)*
EEG & MoBI Analyses	

EDUCATION

University of Florida – Biomedical Engineering August 2020-December 2025
Doctor of Philosophy in Biomedical Engineering, GPA: 3.85
Foci: Neural Engineering and Biomechanics
Human Neuromechanics Laboratory

University of Florida – College of Engineering August 2015-December 2019
Bachelor of Science in Chemical Engineering, GPA: 3.64
Minor in Mathematics
Magna Cum Laude

RESEARCH & EXPERIENCE

Lake-Sumter State College November 2025-Present
Director, Health Professions Simulations
Manage and develop simulations facilities and technology to aid educators of nursing, medical laboratory technician, and respiratory therapy students.

- Meet accreditation standards by creating workflows and standard operating procedures for laboratory use, technology maintenance, and faculty for professional development.
- Integrate audio-visual recording, virtual reality, and high-fidelity manikins into simulations used for training nursing and respiratory therapy students in psychosocial and motor techniques.
- Propose uses of AI agents for developing “soft skills” in patient management and care (e.g., mental health facilities).
- Aid in the design and implementation of new simulations facilities. Coordinate information technology, construction, financial, and academic professionals to build and improve simulation spaces for training nursing and respiratory therapy students.

University of Florida - Biomedical Engineering August 2020-August 2025
Graduate Assistant
Graduate Assistantship under the mentorship of Daniel Ferris Ph.D. in the Human Neuromechanics Laboratory.

- I collaborated with several interdisciplinary labs on a large NIH U01 study aimed at defining the effects of aging on human biomechanics, brain function and structure, and clinical assays (Mind in Motion, NIA 1U01AG061389-01). Our lab measures EEG and gait kinematics while younger and older adults walk on a treadmill. We correlate the

biomechanical and EEG measures to test differences between older and younger adults to better understand decline in mobility that occurs with aging.
- I led EEG data collections for the study. In coordination with 4-6 researchers, I collected EEG, IMU, and ground reaction force data from older, and younger adults.

Best Buy Health & Technology Center June 2021-January 2022
Applied Machine Learning - Graduate Intern
I worked on an agile, multidisciplinary team with full-time employees in engineering, data science, and quality. I gained exposure into the development of medical and wellness devices.
- I tested the sensitivity of a wrist and chest inertial measurement unit to detect falls in older adults. I developed a new measure to detect falls from using a barometer sensor. This measure and other derived measures of falls were included in a random forest model to be implemented on an imbedded fall detection device (Lively Mobile2).
- I improved the machine learning algorithm's ability to detect falls by ~20%

University of Florida – Public Health and Health Professions January 2020-August 2020
OPS Laboratory Technician - Mitchell Lab
I organized the laboratory space, and I performed maintenance on laboratory equipment. I aided with electrophysiological experiments aimed at improving our understanding of respiratory physiology.

University of Florida - Public Health and Health Professions October 2017-January 2020
Undergraduate Research Assistant - Fuller Lab
- I performed animal care and assisted with in-vivo neurophysiology experiments to support experiments on the spinal cord's role in respiration.

University of Florida & City of Gainesville June 2017-July 2018 & May 2019-December 2020
Lifeguard

PEER-REVIEWD JOURNAL ARTICLES

Liu C, Pliner EM, **Salminen J**, Downey RJ, Hwang J, Roy A, Swearinger R, Richer N, Hass CJ, Clark DJ, Manini TM, Cruz-Almeida Y, Seidler RD, & Ferris DP (2025) Age Differences in Electrocortical Dynamics During Uneven Terrain Walking*. Imaging Neuroscience; 3 IMAG.a.1039. doi: https://doi.org/10.1162/IMAG.a.1039*

Salminen J, Liu C, Pliner EM, Tenerowicz M, Roy A, Richer N, Hwang J, Hass CJ, Clark DJ, Cruz-Almeida Y, Manini TM, Seidler RD, & Ferris DP (in revision) Interstride Variation in EEG Power Spectra of Younger and Older Adults Walking at a Range of Gait Speeds. (2026), *IEEE Transactions on Neural Systems and Rehabilitation Engineering; vol. 34, pp. 952-965, 2026, doi: https://doi.org/10.1109/TNSRE.2026.3656061*

Salminen J, Liu C, Pliner EM, Tenerowicz M, Roy A, Richer N, Hwang J, Hass CJ, Clark DJ, Seidler RD, Manini TM, Cruz-Almeida Y, & Ferris DP (2024) Gait Speed Related Changes in Electrocortical Activity in Younger and Older Adults. *Journal of Neurophysiology*, 133(6):1761-1794. doi: https://doi.org/10.1152/jn.00544.2024

Liu C, Downey RJ, **Salminen JS**, Rojas SA, Richer N, Pliner EM, Hwang J, Cruz-Almeida Y, Manini TM, Hass CJ, Seidler RD, Clark DJ, Ferris DP (2024) Electrical Brain Activity during Human Walking with Parametric Variations in Terrain Unevenness and Walking Speed. *Imaging Neuroscience*, 2:1-33. doi: https://doi.org/10.1162/imag_a_00097

ABSTRACTS

DeVol C.R., Liu C., **Salminen J.S.,** Pliner E.M., Roy A., Hass C.J., Clark D.J., Manini T.M., Seidler R.D., Ferris D.P. "Modulation of Broadband EEG During Walking in Younger and Older Adults" (2026). Neural Control of Movement Conference (Poster).

Salminen J.S., Liu C., Pliner E.M., Roy A., Richer N., Hwang J., Clark D.J., Seidler R.D., Hass C.J., Manini T.M., Cruz-Almeida Y., Ferris D.P. "*Interstride Spectral Power in Theta-band EEG is Correlated with Mediolateral Excursion in Both Older and Younger Individuals Walking at a Range of Speeds*" (2025). International Society for Gait and Posture Conference (Podium Presentation).

Tasci S., Sato S.D., **Salminen J.S.,** Clark D.J., Ferris D.P., Hass C.J., Manini T.M., Seidler R.D. "Uneven Terrain Walking is Associated with Brain White Matter Characteristics in Young and Older Adults with Varying Physical Function" (2025). International Society for Gait and Posture Conference (Podium Presentation).

DeVol C.R., Liu C., Pliner E.M., **Salminen J.S.,** Hass C.J., Clark D.J., Manini T.M., Seidler R.D., Ferris D.P. "Association between Mobility Function and Aperiodic EEG Components in Older Adults" (2025). International Society for Gait and Posture Conference (Podium Presentation).

Salminen, J.S., Liu, C., Pliner, E.M., Richer, N., Hwang, J., Ferris, D.P. "*Electrocortical Dynamics of Older Adults Walking at Different Speeds and on Varied Terrain Difficulties*" (2024). Mobile Brain/Body Imaging (MOBI) Conference (Podium Presentation).

Pliner E.M., Liu C., **Salminen J.S.**, Richer N., Hwang J., Clark D.J., Seidler R.D., Hass C.J., Manini T.M., Cruz-Almeida Y., Ferris D.P. "Compensation Related Utilization of Neural Circuits (CRUNCH) of Electrocortical Activity during Walking on Terrain Unevenness" (2024) Mobile Brain/Body Imaging (MOBI) Conference. (Podium Presentation).

Liu, C., Downey, R.J., **Salminen, J.S.,** Ferris, D.P. "Neural Oscillations During Uneven Terrain Walking" (2023). IEEE Neural Engineering Conference Abstract (Podium Presentation).

Streeter K.A., Wollman L.B., Fusco F.A., Poirier T.P., **Salminen J.S.,** Fuller D.D.,"Electrical activation of phrenic afferent neurons evokes phrenic motor plasticity following cervical spinal cord injury", (2019) Experimental Biology Conference Abstract (Podium Presentation).

LEADERSHIP & INVOLVEMENT

University of Florida Counselling & Wellness Center **January 2023-Present**

- Collaborate in support with students that are struggling with emotional turmoil affecting their lives.

University of Florida, Biomedical Engineering Department August 2021-May 2024

Girls with Nerve, Member

- Instruct middle school children from local schools about neuroscience, neural engineering, and research in related fields.

American Society of Biomechanics, Member

- Biomechanics Day: teach children from local schools about the effects of virtual reality on balance in a high stress environment, like walking on a tightrope.

WORKSHOPS & CONFERENCES

International Society for Gait and Posture World Congress; Maastricht, Netherlands (2025)
Mobile Brain Body Imaging Workshop & Conference; *Piran, Slovenia* (2024)
University of Florida, Biomedical Engineering Pruitt Research Day, Gainesville, FL (2024)
Workshop Toward Open and Robust Neuroimaging; Delmenhorst, Germany & Online (2024)
Mobile Brain Body Imaging Workshop & Conference; *San Diego, CA, USA* (2022)
Neurotherapeutic Intermittent Hypoxia Workshop; *Jacksonville, FL* (2020)

Center for Respiratory Research and Rehabilitation Workshop; Gainesville, FL (2019)

HONORS & AWARDS

International Society for Posture and Gait Research Travel Stipend (2025)
Office of Research Travel Award, University of Florida (2025)
Graduate School Preeminence Award, University of Florida (2020)

GRANTS

NIH F31 (UEI: NNFQH1JAPEP3), "Electrocortical Dynamics of the Aging Brain During Gait". PI: Jacob S Salminen, COPI: Daniel P Ferris. (not funded)
NSF GFRP (#1000322146) "Deep Learning Methods for Capturing Electrocortical Connectivity" (not funded)

INSTITUTIONAL REVIEW BOARDS

(IRB202100978 UF), "Brain Dynamics of Immersive Virtual Reality Training"

TEACHING EXPERIENCE

Quantitative Physiology
Guest Lecturer

- I lectured 3 classes on probability theory, careers in computational neuroscience and biomechanics, and gave an overview of Markov sequences, support vector machines, and k-nearest neighbor classifiers.

Biomedical Instrumentation (Lecture and Lab) January-May 2021 & August-December 2021
Teaching Assistant

- Laboratory portion: I prepared labs to give students experience in developing and evaluating a biomedical device. The final project was to have groups of students build an electrocardiograph.
- Lecture portion: I provided students with guidance on homework, quizzes, and exams. I provided exam preparation lectures with guided problem solving.